Road Accident Pattern Analysis, India (DA-022)

Domain: Road Safety Analytics

Problem: A road-safety team wants to understand when and where recorded accidents in India are more frequent.

Objectives: (1) Summarize accident counts and severity. (2) Compare time periods and conditions. (3) Identify data-quality gaps.

Dataset: Indian Road Accident Dataset (2022-2025), Kaggle. 20,000 records after cleaning (from 20,000).

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Google Colab.

Method: loaded data, removed duplicate ids, parsed dates, checked hours and locations, built weekday/month/period features, built charts.

Note: Descriptive analysis. Correlation does not prove causation and no blame is assigned.

Key Findings

1. Accidents peak at 2:00 and are lowest at 7:00. The busiest time period is Night (0-5).

2. Monday has the most accidents (2,966) and Wednesday the fewest (2,726); weekends hold 28.6% of accidents.

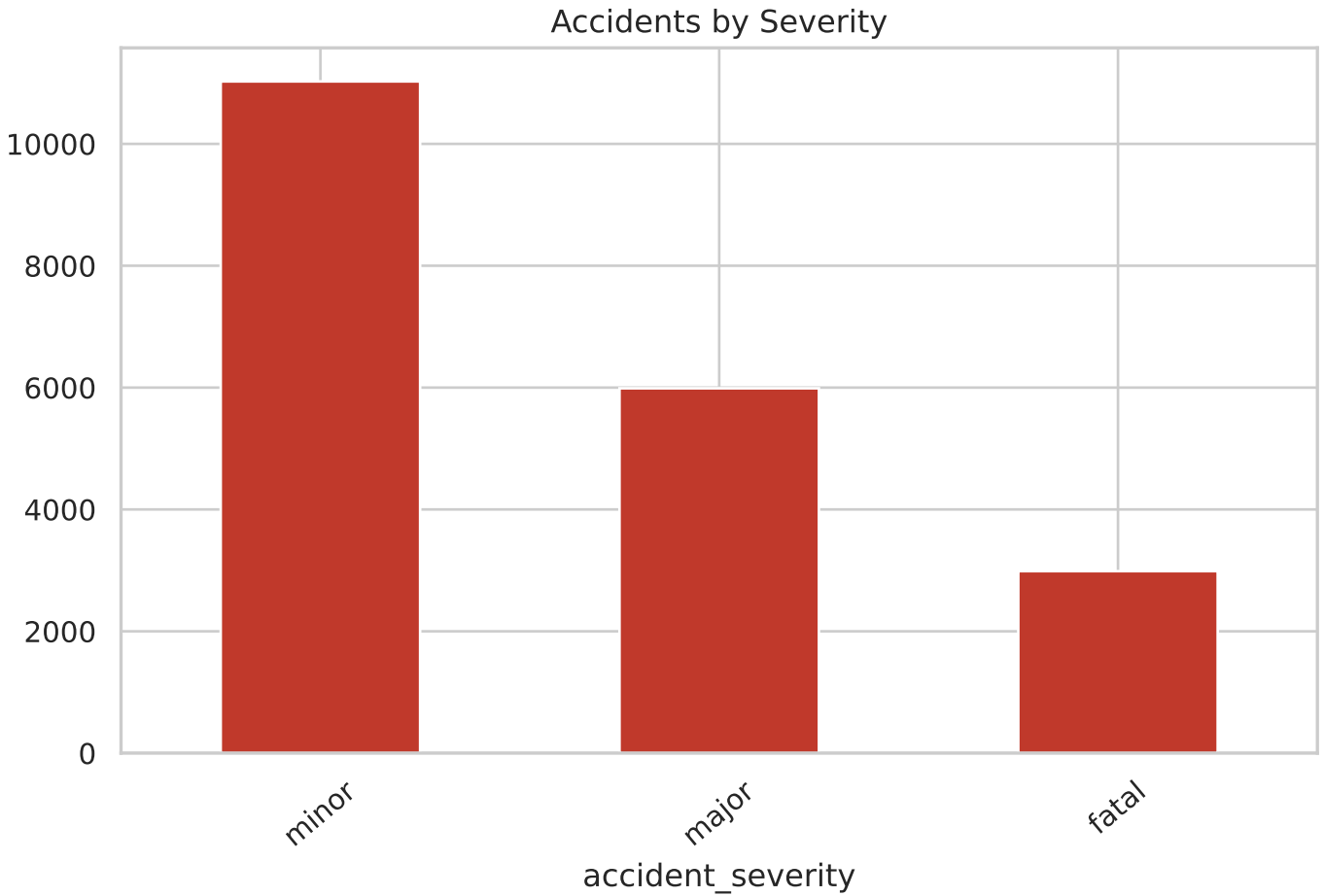
3. 44.9% of accidents are major or fatal; the most common severity is minor.

4. clear weather has the highest share of severe accidents (45.0%). This is an association, not a cause.

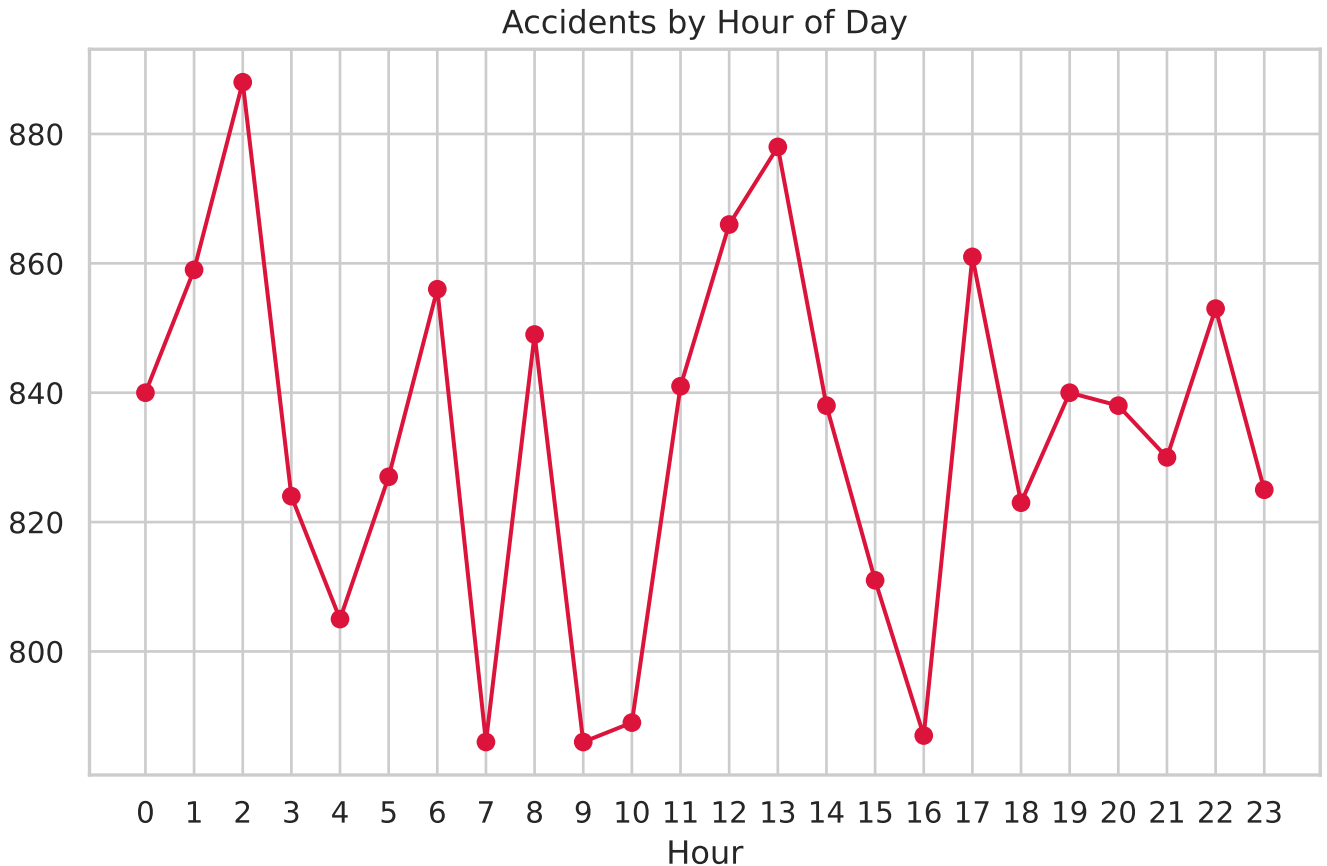
5. The most common recorded cause is distraction (20.1% of accidents).

6. Maharashtra has the most records, and Chandigarh is the top city.

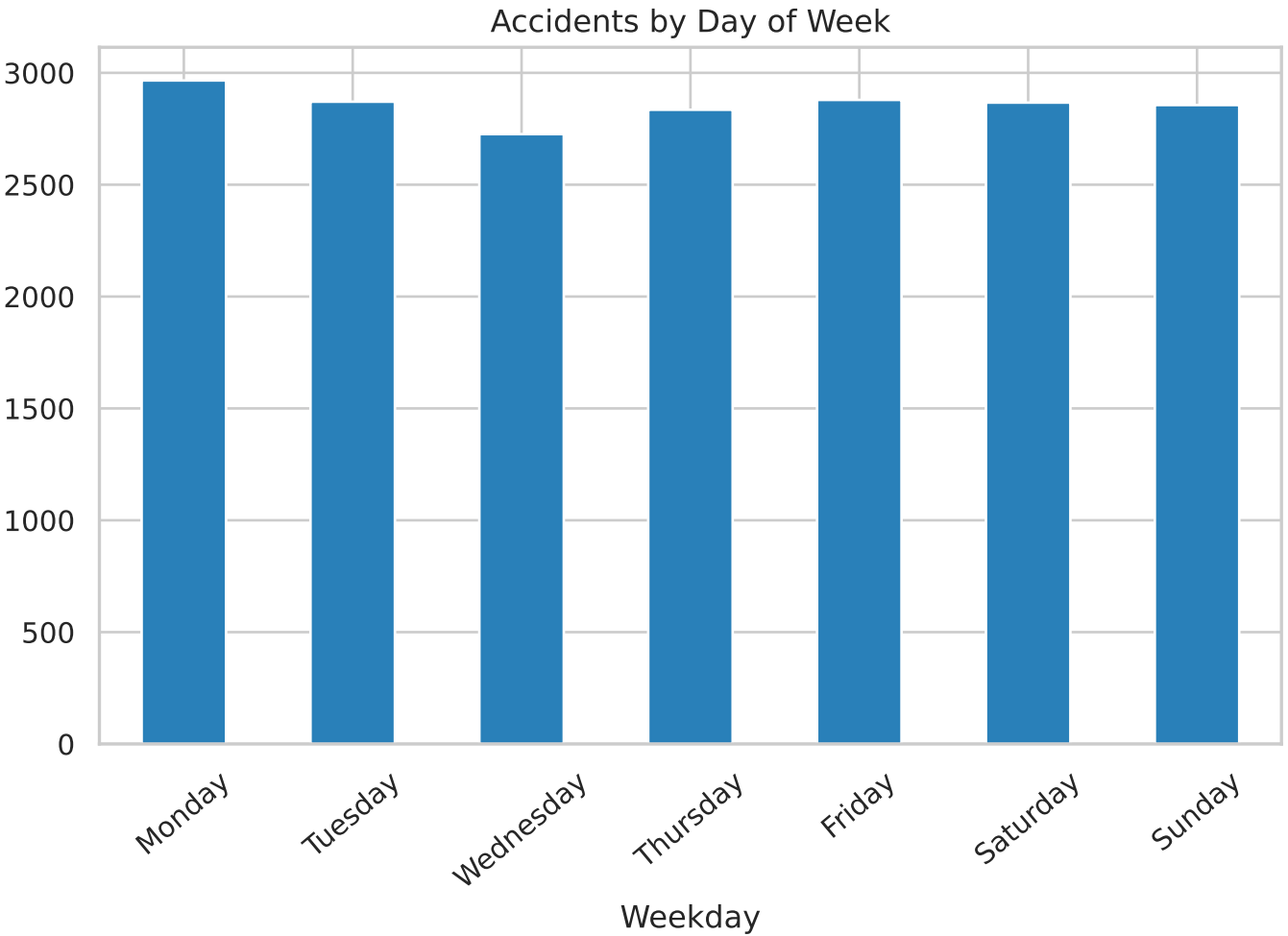
7. Among road features, "traffic_density: medium" has the highest severe share (45.4%).



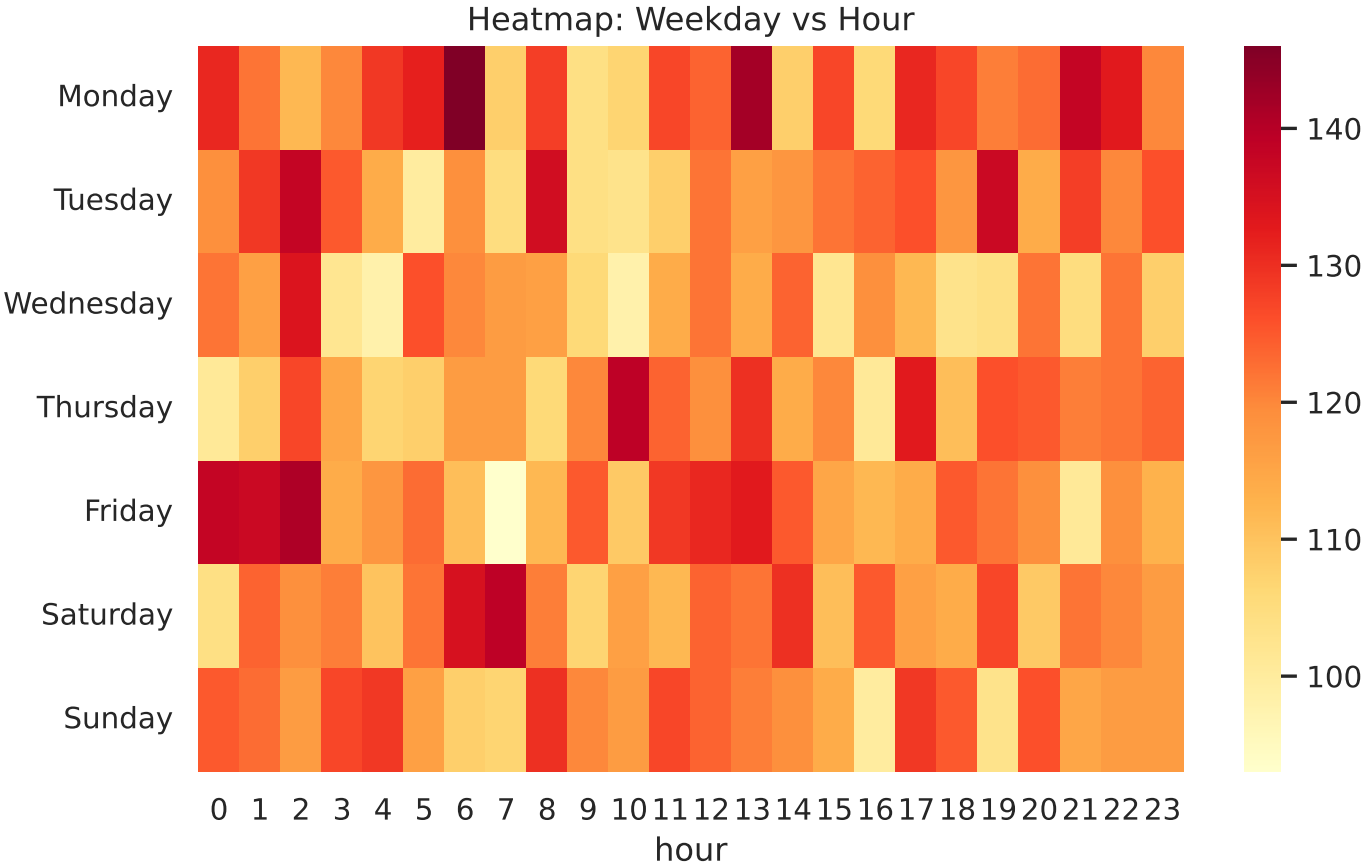
Insight: minor is the most common severity; 44.9% are major or fatal.



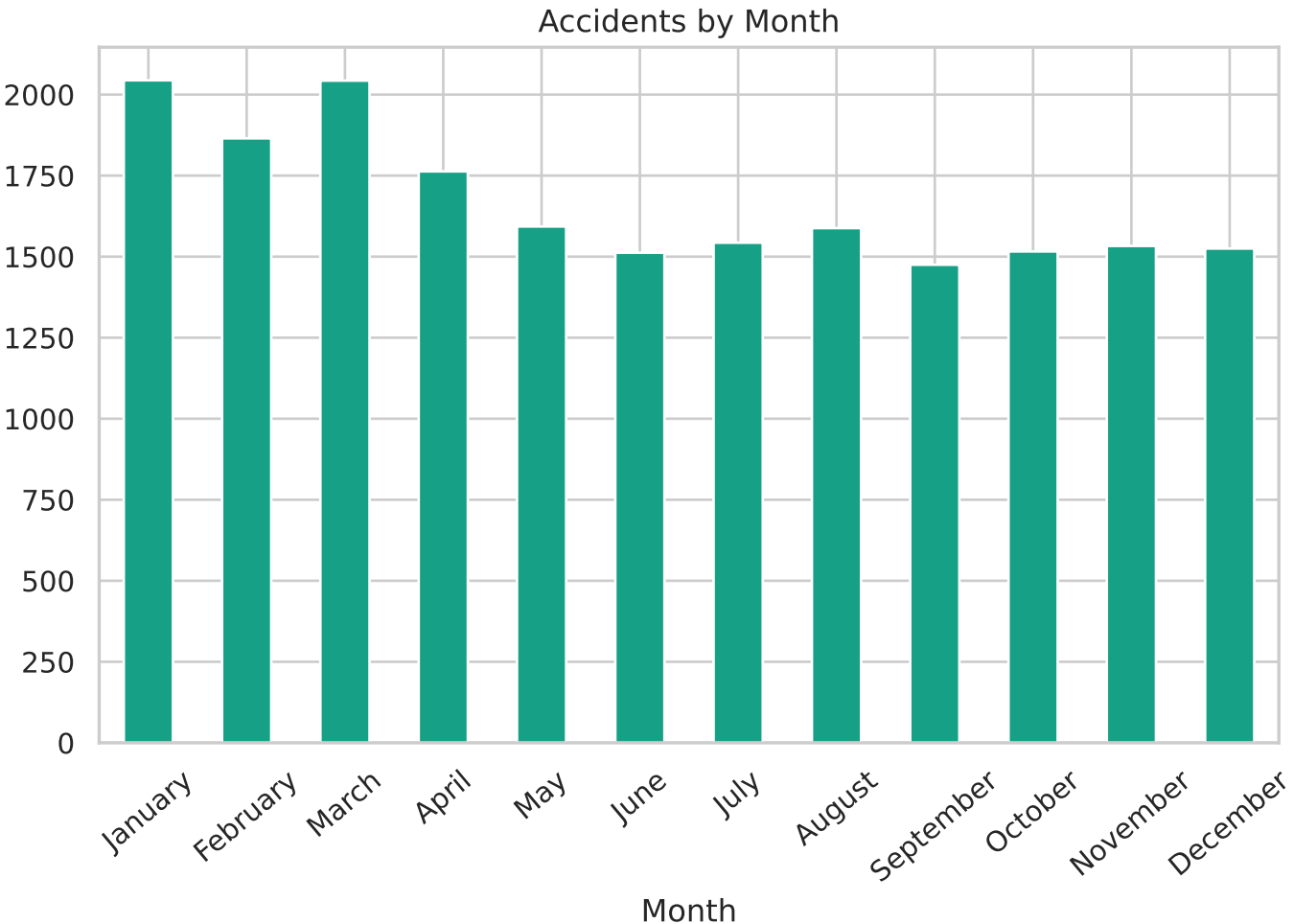
Insight: Peak at 2:00; lowest at 7:00.



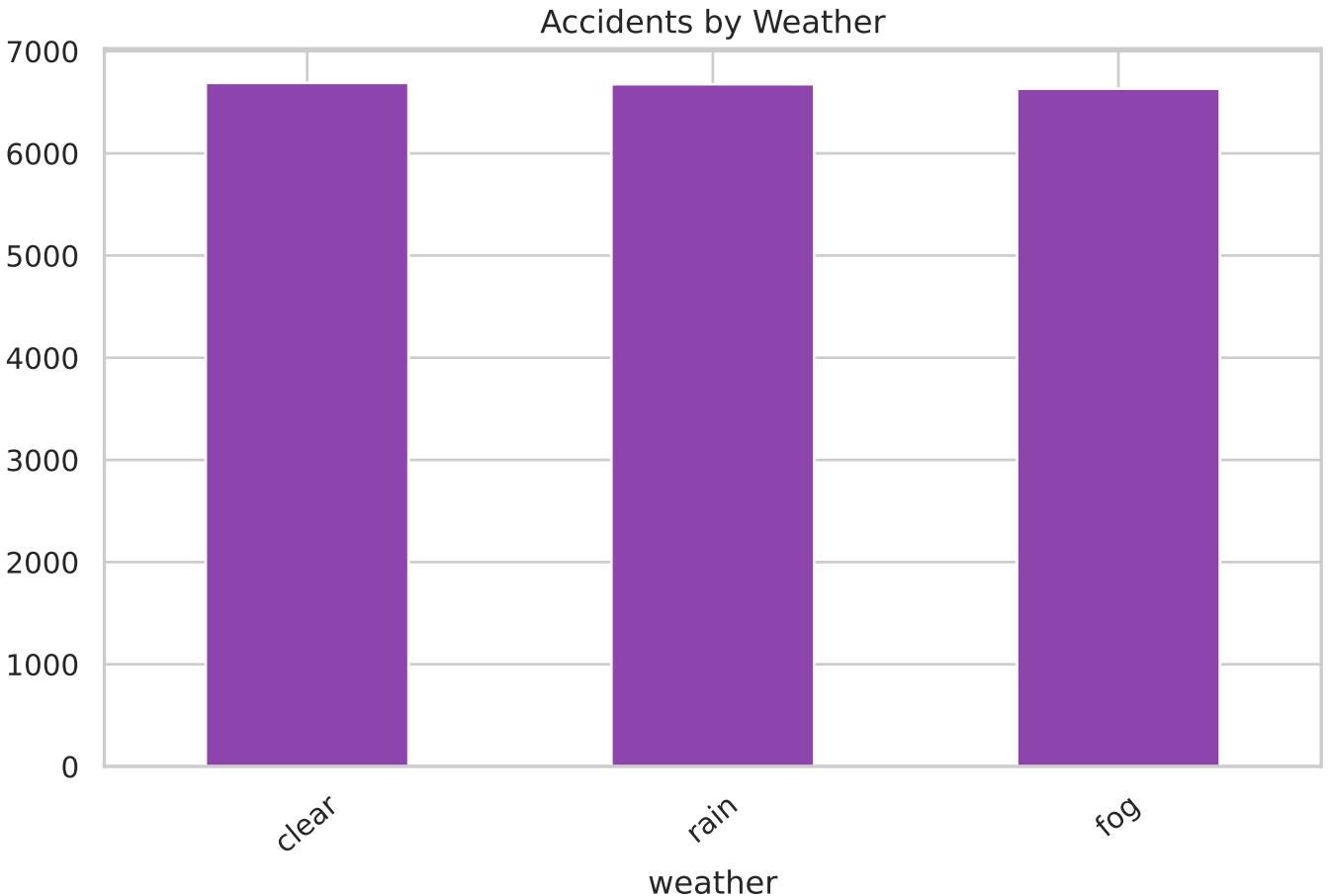
Insight: Monday highest, Wednesday lowest.



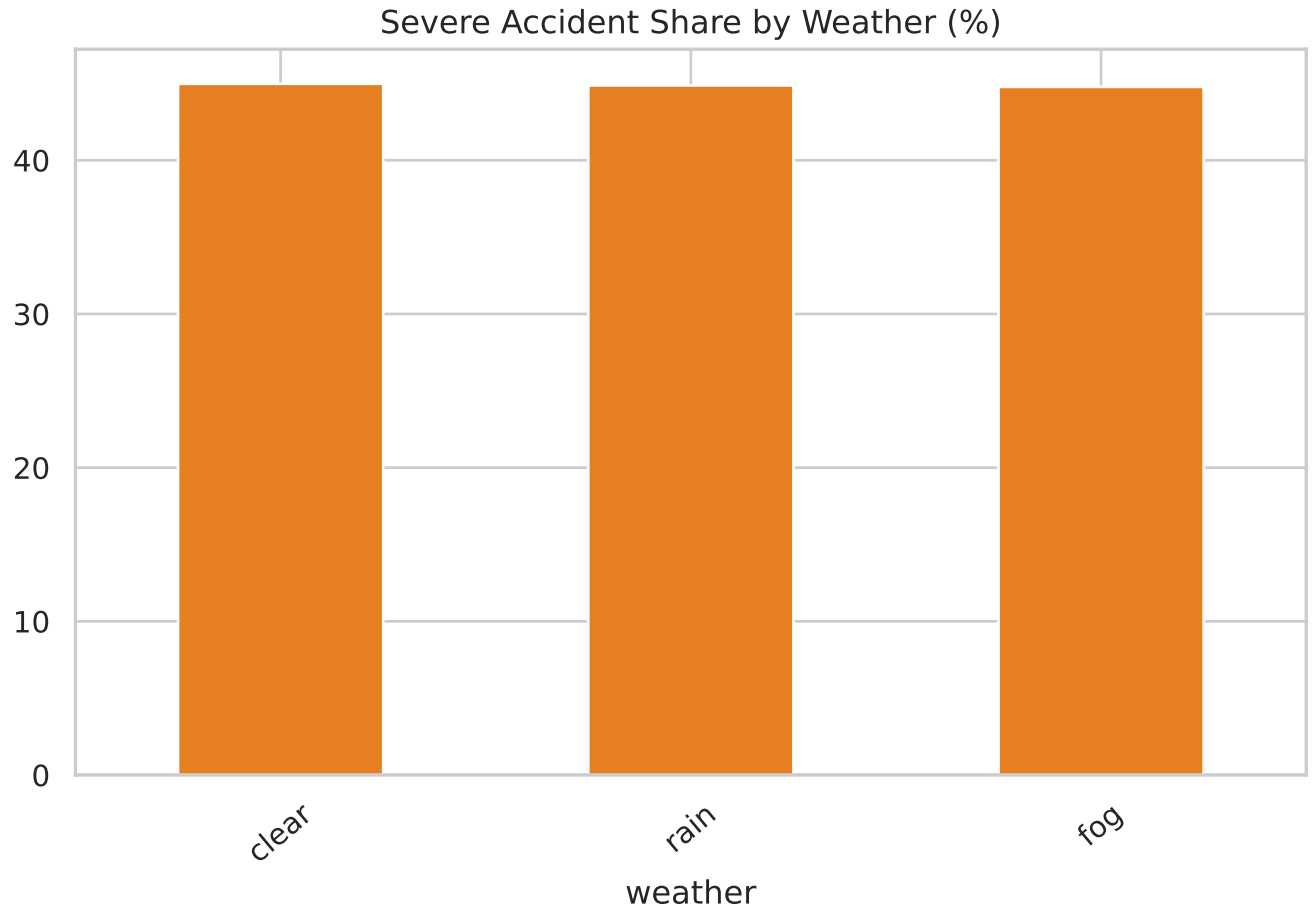
Insight: Darker cells mean more accidents.



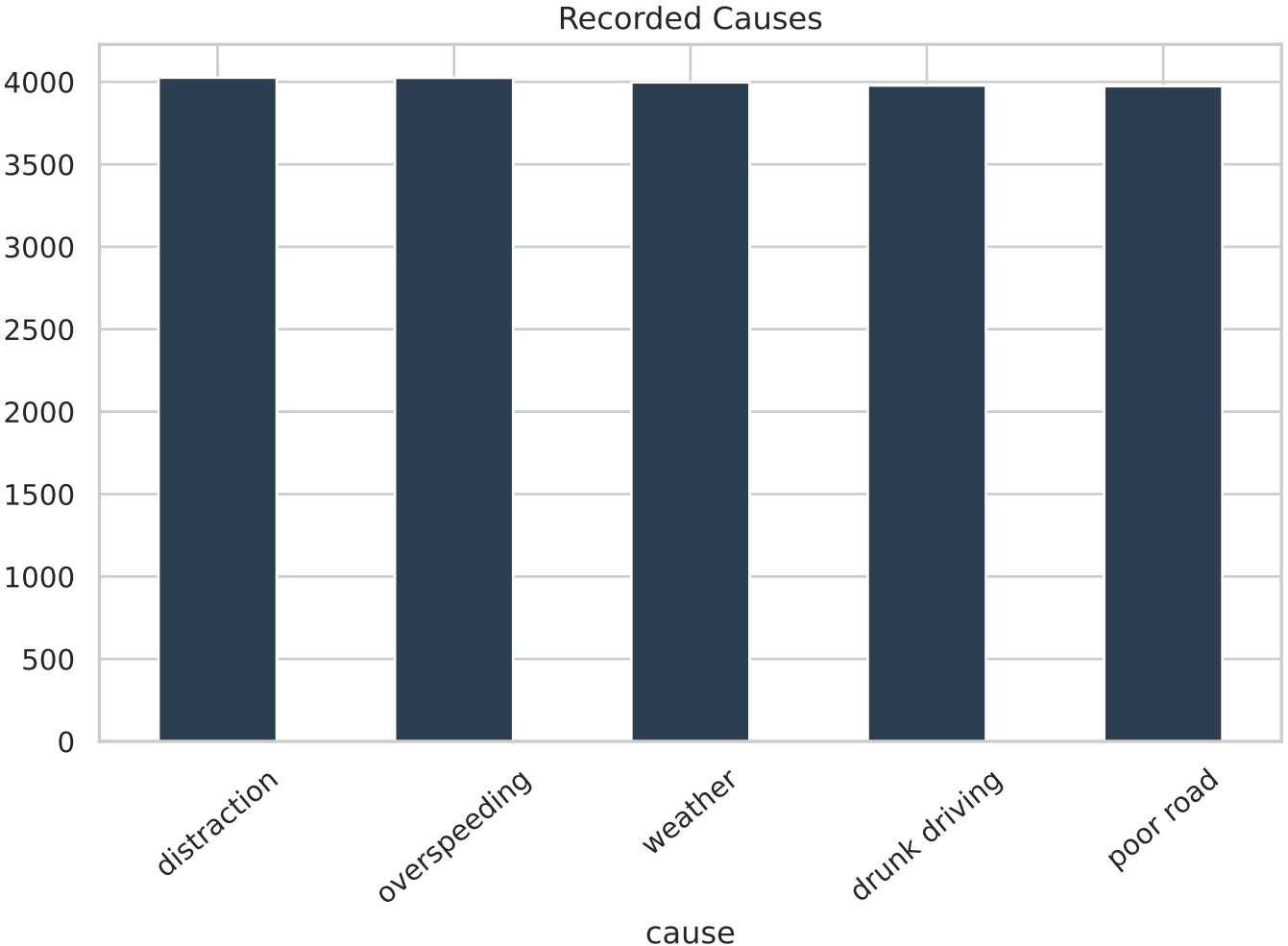
Insight: January has the most accidents in this data.



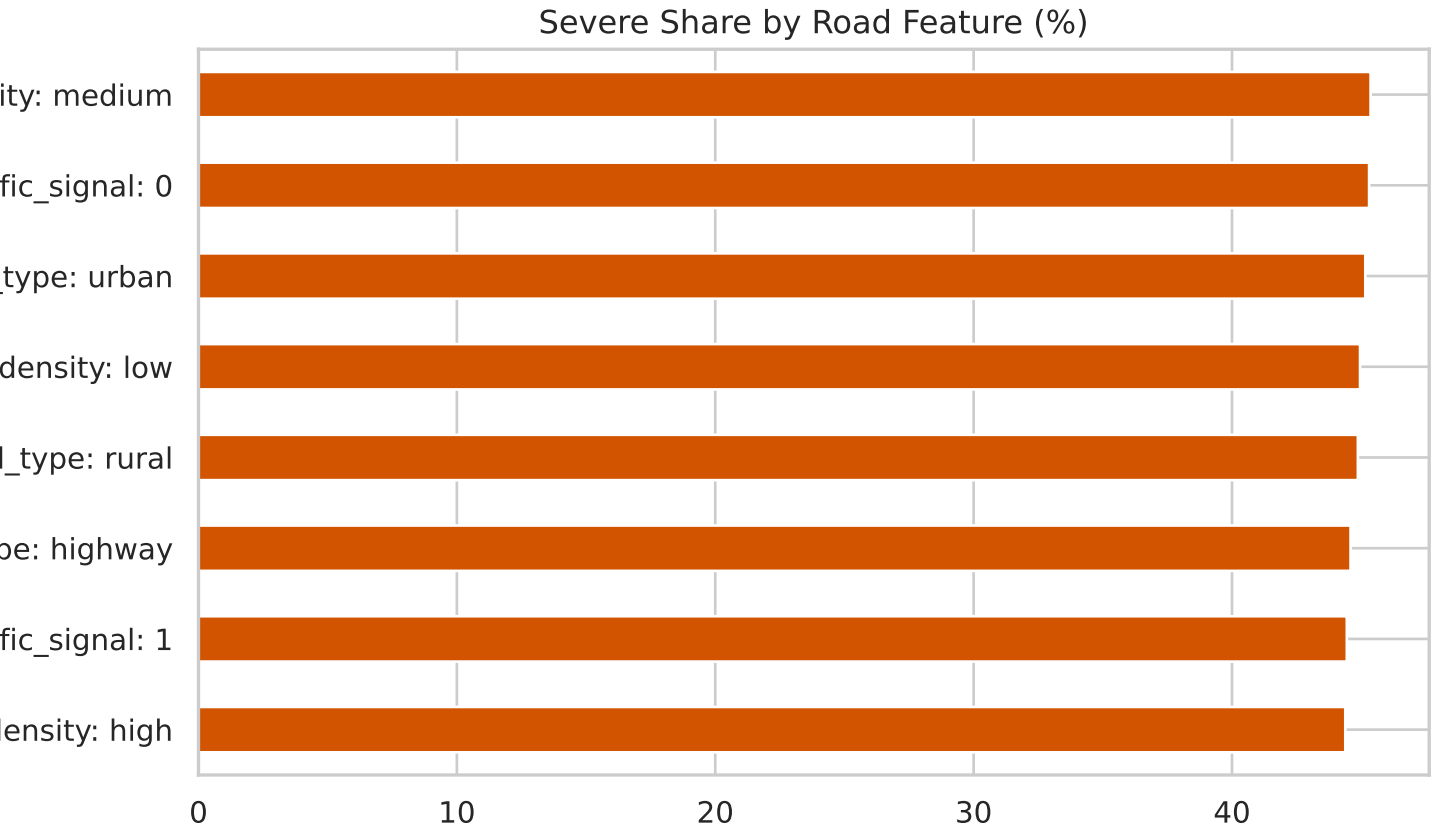
Insight: clear has the most accidents because it is the most common condition.



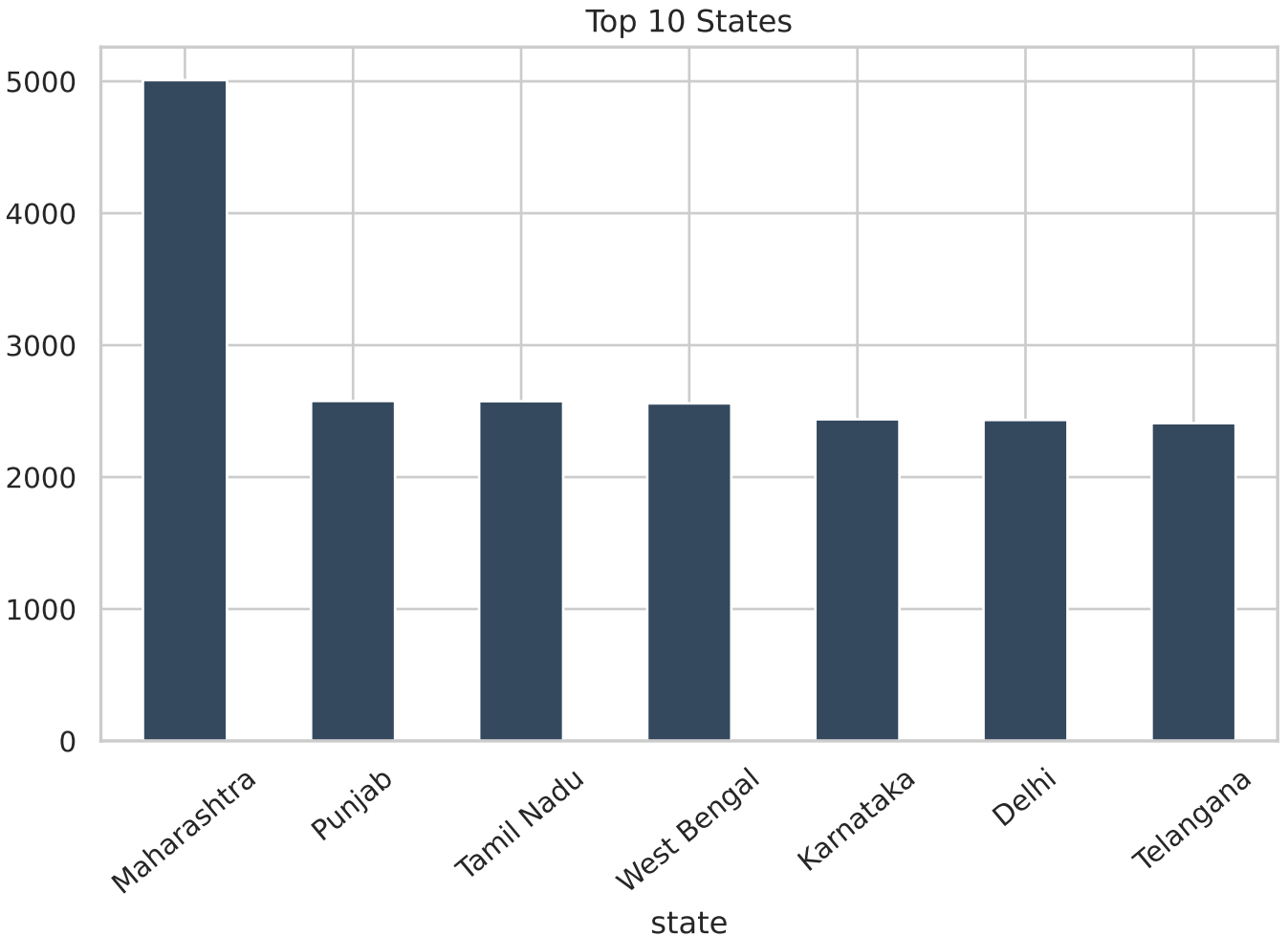
Insight: clear has the highest severe share (45.0%). Association, not cause.



Insight: distraction is the most common recorded cause.



Insight: "traffic_density: medium" has the highest severe share.



Insight: Maharashtra has the most records.

Data Quality, Limitations and Conclusion

Rows in file: 20,000. Duplicate ids: 0. Invalid dates: 0. Invalid hours: 0. Locations outside India bounds: 0.

Columns with missing values: festival (99.42%)

Limitations: the collection method of the Kaggle dataset should be verified; counts are not adjusted for traffic volume; some columns are mostly empty.

Findings show associations, not causes.

Conclusion: This analysis of 20,000 recorded accidents in India shows clear time and place patterns. Accidents peak at 2:00, with the busiest period being Night (0-5), and Monday has the most accidents. 44.9% of accidents are major or fatal, and the most common recorded cause is distraction. Maharashtra has the most records. These are patterns only and do not prove causes. Questions for further study: why do the peak hours stand out, do high-severity road features need better safety measures, and do the same patterns hold in official, full-year data?